

Self-Monitoring through Rhythm: Probing Prosodic Planning with Delayed Auditory Feedback

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Background

THE QUESTION: What is driving the speech disruptions under delayed auditory feedback (DAF)?

Two competing theories: **SEGMENTAL** vs. **SUPRASEGMENTAL**

The Evidence

- Max disruption at **~200ms delay** aligns with **syllabic** timing (MacKay, 1968).
- Impact from **stress patterns** > **phonetic content** (Kaspar & Rübeling, 2011).

However, there is no direct test of prosody vs. segments so far.

Research Question & Hypothesis

This pilot study investigates the primary driver of DAF-induced speech disruption.

Guiding Question:

Does reducing prosodic variation (i.e., speaking monotonously) mitigate speech disruption under delayed auditory feedback?

Hypothesis:

We predict that DAF will cause significantly greater disruption during natural, prosodically rich speech compared to monotonous speech.

This will suggest that DAF interferes with the **temporal tracking of prosodic frames**, not just segmental articulation.

Topic Areas:

Speech Motor Control, Language Production

Analysis Criteria

Speech recordings will be analyzed for three primary behavioral metrics:

Disfluencies: Frequency of repetitions and prolongations.

Speech Rate: Calculated in syllables per second.

Error Patterns: Mistimed syllables relative to prosodic structure.

Pilot Methods

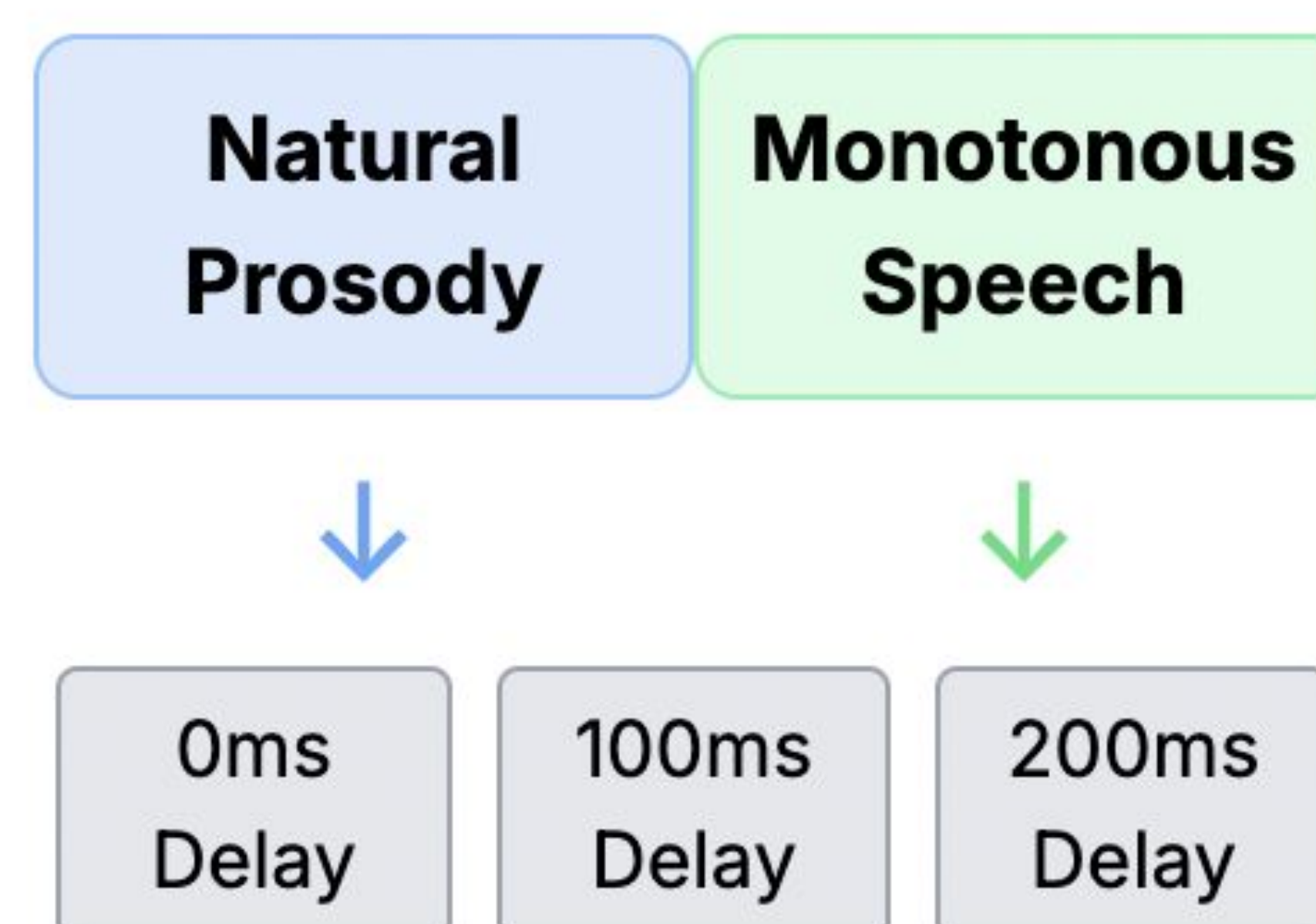
Participants

5 adult English speakers (F = 4) with no history of speech, language, or hearing disorders.

Experimental Design

A **2 × 3 within-subjects design** was employed to compare Speech Production Mode across three Feedback Delay conditions.

2×3 Design Visualization



Stimuli & Tasks

Participants read from balanced sentences, chants, and poetry. Here is an example of two contrastive tasks used:

1. Sentence Prosody Contrast

Normal Prosody

"I do **NOT** wish to fall!"

Monotonous

"I do not wish to fall."



2. Counting Task Contrast

Segmental

"one two"

Prosodic

"**ONE** one"

Expected Results

We expect a significant interaction between Speech Mode and Delay.

Disruption is predicted to be **greatest at the 200ms delay**, with this effect being substantially more pronounced in the **Natural Prosody** condition.

Implications

This study aims to clarify the relationship between prosodic frames and monitoring of speech timing.

Clinically, this work could point to new directions for temporal processing interventions for speech disorders, such as stuttering, for prosodic and rhythmic-based therapies.

References

Kaspar, K., & Rübeling, H. (2011). Rhythmic versus phonemic interference in delayed auditory feedback. *JSLHR*, 54(3), 932–943.

MacKay D. G. (1968). Metamorphosis of a critical interval: age-linked changes in the delay in auditory feedback that produces maximal disruption of speech. *JASA*, 43(4), 811–821.